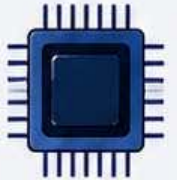
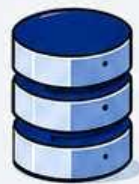
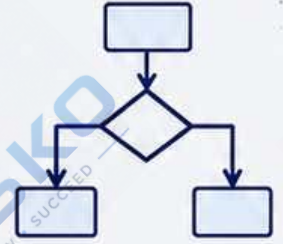


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# Syllabus





**Teaching and Learning Scheme: for the Degree of Bachelor of Science with the Major: Computer Science/ Information Technology/ Computer Application (Regular)/ Computer Application (Vocational)/ Data Analytics**

**(Three Years- Six Semesters Bachelor's Degree Programme)**

**(Four Years- Eight Semesters Bachelor's Degree Programme (Honors))**

**(Four Years- Eight Semesters Bachelor's Degree Programme (Honors with Research))**

**Preamble**

The new curriculum of the four-year undergraduate program under NEP, for Computer Science aims to develop the core competence in computing and problem solving amongst its graduates. Informally, "Learning to learn" has been the motto of the department since its inception. The curriculum thus focuses on building theoretical foundations in Computer Science to enable its pupils to think critically when challenged with totally different and new problems. It imbibes the following **Student-Centric** features of NEP2020:

**Flexibility to Exit:**

In order to support early exits, the curriculum aims to develop employability skills early. This has been done so that the outcomes of the 4 yr degree is not compromised as we believe that all but a few students will go for the full 4-year degree. As programming is at the heart of computing it is proposed to have two programming courses early so that the students can develop good programming skills in the first year. At the same time students are familiarized with the hardware of computers early on.

**Employability:**

Industry demand in the IT sector has changed considerably in the past few years. With the humongous amount of data coming from all the domains like medical data, social networking data, astronomical data, education, etc., automating information extraction and analysis of data is the only way forward to leverage the available data for the future. The curriculum aims to equip the students with tools and techniques of Artificial Intelligence, Machine Learning and a pathway on Data Science if the student so desires. Having said this, there is no replacement for the foundational courses like programming, data structures and algorithms. With two courses on programming and three courses on data structures and algorithms together, a strong foundation will be laid down for problem solving.

**Research:**

With the option to obtain specialization in an area of their choice, the curriculum prepares the students to take up research projects in their final year.

**Program Outcomes:**

**Knowledge outcomes:** After completing B.Sc. Computer Science Program students will be able to:

PO1: To develop problem solving abilities using a computer.;

PO2: To prepare necessary knowledge base for research and development in Computer Science.

**Skill outcomes:** After completing B.Sc. Computer Science Program students will be able to:

PO3: To build the necessary skill set and analytical abilities for developing computer-based solutions.

PO4: To train students in professional skills related to Software Industry.

**Generic outcomes:** Students will

PO5: Augment the recent developments in the field of IT and relevant fields of Research and Development.

PO6: Enhance the scientific temper among the students so that to develop a research culture and Implementation the policies to tackle the burning issues at global and local level.

### **Program Specific Outcomes**

PSO1: Students get knowledge and training of technical subjects so that they will be technical professional by learning C programming, Relational Database Management, Data Structure, Software Engineering, Graphics, Java, PHP, Networking, Theoretical Computer Science, System programming, Object Oriented Software Engineering.

PSO2: Students understand the concepts of software application and projects.

PSO3: Students understand the computer subjects with demonstration of all programming and theoretical concepts with the use of ICT.

PSO4: Development of in-house applications in terms of projects

PSO5: Students will build up programming, analytical and logical thinking abilities.

PSO6: Aware them to publish their work in reputed journals

PSO7: To make them employable according to current demand of IT Industry and responsible citizen.



| Level | Semester | Course Code                                  | Course Name                     | Credits | Teaching Hours | Exam Duration | Max Marks |
|-------|----------|----------------------------------------------|---------------------------------|---------|----------------|---------------|-----------|
| 4.5   | I        | 109200/ 110200/<br>112200/ 123200/<br>134200 | <b>Fundamentals of Computer</b> | 2       | 30             | 2 Hrs         | 30        |

|                           |                                                                                                                                                                                                                                                                                                                                                                             |                          |                                    |                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course Objectives:</b> | 1. To provide the knowledge of basic of Computer Science<br>2. To understand importance of memory devices of computer<br>3. To understand importance of memory devices input output devices of computer<br>4. To understand the Operating System concepts                                                                                                                   |                          |                                    |                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Course Outcomes:</b>   | Students will be able to -<br>1. Define Computer, History of Computer, Uses of Computer and Generations of Computers.<br>2. Define memories of computers, its types and examples of primary and secondary memories.<br>3. Introduce about all input-output devices of computer systems.<br>4. Define operating system, its function and types of operating systems.         |                          |                                    |                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Unit System</b>        | <b>Contents</b>                                                                                                                                                                                                                                                                                                                                                             | <b>Workload Allotted</b> | <b>Weightage of Marks Allotted</b> | <b>Incorporation of Pedagogies</b>                                                                                                                                                                                                                                                                                                                                                         |
| <b>Unit I</b>             | Introduction to Computer, Uses of Computers, History of Computers, Characteristics, Generations of Computers, Block diagram of Computer.                                                                                                                                                                                                                                    | 8 Hrs                    | 8 Marks                            | BoS shall recommend suitable pedagogical strategies, both classical and contemporary innovations, for integration into the Teaching, Learning, and Evaluation (T, L, & E) Processes. These strategies should be tailored to enhance the delivery and comprehension of the course content within each Unit, ensuring that they align with the educational objectives and learning outcomes. |
| <b>Unit II</b>            | <b>Memories:</b> Primary Memories: RAM, ROM, and its types, Cache Memory, Secondary Storage Devices: Hard Disk, SSD, Pen drives.                                                                                                                                                                                                                                            | 7 Hrs                    | 7 Marks                            |                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Unit III</b>           | <b>I/O Devices:</b> Input Devices- Keyboard, Mouse, Scanner, Output Devices- Touch Screen, Monitors: VDU, LCD & LED. Printers: Types of Printers, Impact and non-impact printers, Modem.                                                                                                                                                                                    | 8 Hrs                    | 8 Marks                            |                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Unit IV</b>            | <b>Operating System:</b> Definition, Functions of Operating System, Types: Batch Mode, Multiprogramming, Timesharing, Online Real Time, Distributed O.S. Booting Process.                                                                                                                                                                                                   | 7 Hrs                    | 7 Marks                            |                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>References :</b>       | <b>Course Material/Learning Resources</b><br><b>Text books:</b><br>1) Computer Fundamentals & Networking - P.K.Sinha<br>2) Fundamentals of Computer - B.Ram<br>3) Internet Book - Clstenes<br>4) Information Technology - Alexies & Mathews - Vijay Nikole<br><b>Reference Books:</b><br>1) Fundamentals of Computer - V.Rajaraman<br>2) Computer Network-Andrew Tennanbaum |                          |                                    |                                                                                                                                                                                                                                                                                                                                                                                            |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p><b>Weblink to Equivalent MOOC on SWAYAM if relevant:</b></p> <ul style="list-style-type: none"> <li>• <a href="https://onlinecourses.swayam2.ac.in/cec19_cs06/preview">https://onlinecourses.swayam2.ac.in/cec19_cs06/preview</a></li> <li>• <a href="https://onlinecourses.swayam2.ac.in/nou20_cs03/preview">https://onlinecourses.swayam2.ac.in/nou20_cs03/preview</a></li> <li>• <a href="https://www.classcentral.com/course/swayam-computer-fundamentals-13950">https://www.classcentral.com/course/swayam-computer-fundamentals-13950</a></li> <li>• <a href="https://onlinecourses.nptel.ac.in/noc19_cs42/preview">https://onlinecourses.nptel.ac.in/noc19_cs42/preview</a></li> <li>• <a href="https://onlinecourses.swayam2.ac.in/aic20_sp06/preview">https://onlinecourses.swayam2.ac.in/aic20_sp06/preview</a></li> <li>• <a href="https://onlinecourses.swayam2.ac.in/cec20_cs02/preview">https://onlinecourses.swayam2.ac.in/cec20_cs02/preview</a></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>Model Questions:</b></p> | <p><b>Short Type (At least 8)</b></p> <ol style="list-style-type: none"> <li>1. What is Computer? Explain its characteristics.</li> <li>2. Explain the history of computer.</li> <li>3. What is the function of memory? What are its types?</li> <li>4. Enlist Input-Output devices of computers.</li> <li>5. What are the types of computers?</li> <li>6. What is the function of Printer?</li> <li>7. What is the Function of Operating System?</li> <li>8. What is Booting Process?</li> </ol> <p><b>Long Type (At least 4)</b></p> <ol style="list-style-type: none"> <li>1. Draw and explain the block diagram of computer.</li> <li>2. Explain the generations of computers.</li> <li>3. What is Memory? Explain its types.</li> <li>4. Explain the types of printers.</li> <li>5. Explain any three Input/ Output devices of Computers.</li> <li>6. Explain the types of operating system.</li> <li>7. Explain the characteristics of computers.</li> <li>8. Explain the uses of computers.</li> </ol> <p><b>MCQs for Internal Assessment (At least 8)</b></p> <ol style="list-style-type: none"> <li>1. Who is the father of Computers? <ol style="list-style-type: none"> <li>a) James Gosling</li> <li>b) Charles Babbage</li> <li>c) Dennis Ritchie</li> <li>d) Bjarne Stroustrup</li> </ol> <p>Answer: b) Charles Babbage</p> </li> <li>2. What is the full form of CPU? <ol style="list-style-type: none"> <li>a) Computer Processing Unit</li> <li>b) Computer Principle Unit</li> <li>c) Central Processing Unit</li> <li>d) Control Processing Unit</li> </ol> <p>Answer: c) Central Processing Unit</p> </li> <li>3. Which of the following is the brain of the computer? <ol style="list-style-type: none"> <li>a) Central Processing Unit</li> <li>b) Memory</li> <li>c) Arithmetic and Logic unit</li> <li>d) Control unit</li> </ol> <p>Answer: d) Control unit</p> </li> <li>4. Which of the following is the smallest unit of data in a computer? <ol style="list-style-type: none"> <li>a) Bit</li> </ol> </li> </ol> |



b) KB  
c) Nibble  
d) Byte  
Answer: a) Bit

5. Which of the following is designed to control the operations of a computer?  
a) User  
b) Application Software  
c) System Software  
d) Utility Software  
Answer: c) System Software